AARON ELLIS

Staff full-stack engineer — TypeScript, React, Node. 12 years of consumer fintech.

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EXPERIENCE

Staff Software Engineer — Topstep

Trading Platform — Charts
Oct 2026 — Present

Staff engineer owning the TopstepX charting engine — a framework-agnostic rendering library that powers real-time charts for thousands of active traders.

- Owns core charting-engine architecture: layer system, hit testing, dirty-region rendering, and the public API that React, React Native, and direct consumers integrate against.
- Sets the 60fps performance bar — frame-budget discipline, Web Workers + SharedArrayBuffer data pipelines, and pointer responsiveness under continuous data flow.
- Defines visual-testing standards for the library: pixel diffing, deterministic replay, and golden-image regression.

Staff Software Engineer — NuvaLabs

Jun 2024 — May 2026

Staff engineer on security-critical real-world-asset fintech: built nvPRIME end-to-end and hardened the platform's authentication and AML enforcement.

- Designed and shipped nvPRIME end-to-end, across frontend, services, and infrastructure.
- Hardened authentication across Auth0 and Privy — session policy, token handling, and the account-linking edge cases where breaches live.
- Implemented AML enforcement controls where a mistake costs real money — compliance logic treated as production code, with tests to match.
- Introduced multi-agent AI development workflows to the team, raising throughput without lowering the review bar.

Full-Stack Engineer (Contract) — Empirium

Jun 2023 — May 2024

Contract full-stack: shipped an escrow.com checkout integration end-to-end and an organization admin portal with role-based access control.

- Built the escrow.com checkout end-to-end — API integration, payment state handling, and the customer-facing flow.
- Delivered the org admin portal: user management, roles and permissions (RBAC), designed to be audit-friendly from day one.
- Operated as a one-person full-stack team inside a client codebase: scoped it, shipped it, handed it off clean.

Senior Software Engineer — Public.com

Apr 2021 — May 2023

Built the bond screener for an industry-leading fractional investing platform, and led the Core Web Vitals work behind measured funnel gains.

- Designed and shipped the bond screener end-to-end: instrument filtering, yield presentation, and the entry points into the buy flow.
- Led the performance effort on the highest-traffic funnel pages — LCP and interaction latency down, with the gains verified in production analytics.
- Tied performance budgets to funnel metrics in CI, so regressions failed builds instead of showing up a quarter later.

Frontend Engineer, SoFi Invest — SoFi

Jun 2017 — Mar 2021

Primary frontend engineer for SoFi Invest; led the Angular → React migration whose architecture was adopted across the organization.

- Carried the Invest web codebase through a full Angular → React rewrite while the product kept shipping — incremental route cutover, no feature freeze.
- The migration architecture — module boundaries, shared design-system consumption, staged rollout — became the template other SoFi teams adopted org-wide.
- Built and maintained investing flows used by millions of members: brokerage onboarding, trading, and automated investing.

SKILLS

Frontend: TypeScript, React, Angular, Web performance (Core Web Vitals), Tailwind CSS, Canvas rendering, Web Workers & SharedArrayBuffer, Rendering performance (60fps budgets)

Backend: Node.js, API design, Payments & escrow integrations, RBAC & permissions, Auth hardening (Auth0, Privy), AML & compliance systems

Cloud: CI/CD (GitHub Actions), Cloudflare Pages & Workers

Leadership · AI: Multi-agent AI dev workflows, Technical leadership & mentoring

EDUCATION & PERSONAL PROJECTS

- 2014–17 — My Menu Plans — built, ran, and sunset a meal-planning product solo; WordPress → React/Node
- 2016 — freeCodeCamp — Full Stack certification
- 2017 — V School — full-stack immersive